



Link Between a Patient's Diet and Cognitive Function

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Abstract

Aging is associated with cognitive decline and increased dementia risk, yet the modifiable factors that drive this, like diet and metabolic health, remain poorly understood at the individual level.

Research Goal:

We are analyzing longitudinal patient data to investigate relationships between:

- Cognitive performance
- Blood biomarkers
- Dietary variables
- Dementia-related outcomes

Main Objective:

Understanding how missing data patterns may affect analysis, correlations, and predictive modeling

Dataset Overview

Studies:

- HCA
- AABC

Data Types:

- Cognitive measurements
- Blood biomarker data
- Dietary intake data
- Patient visit/event history

Longitudinal Design:

- Patients across visits, events, and years (some followed up to 8 years)

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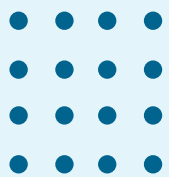
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01

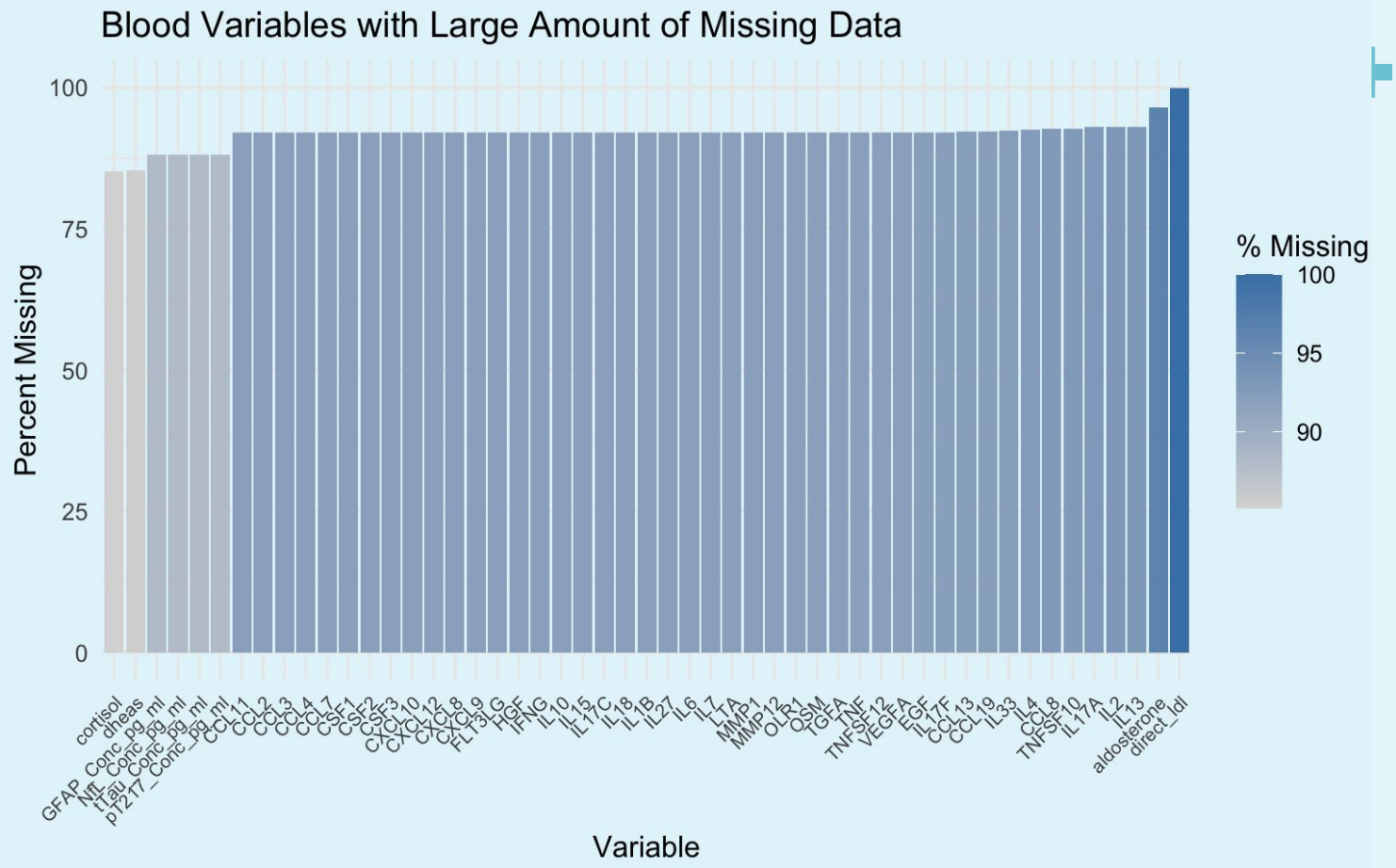
Missingness Report

A great deal of data is missing (NA) – therefore our data is heavily **biased**. We would like to determine in *what way* its **biased**. Perhaps by adjusting for this bias we see strong associations for each variable to the cognitive markers in studying dementia.



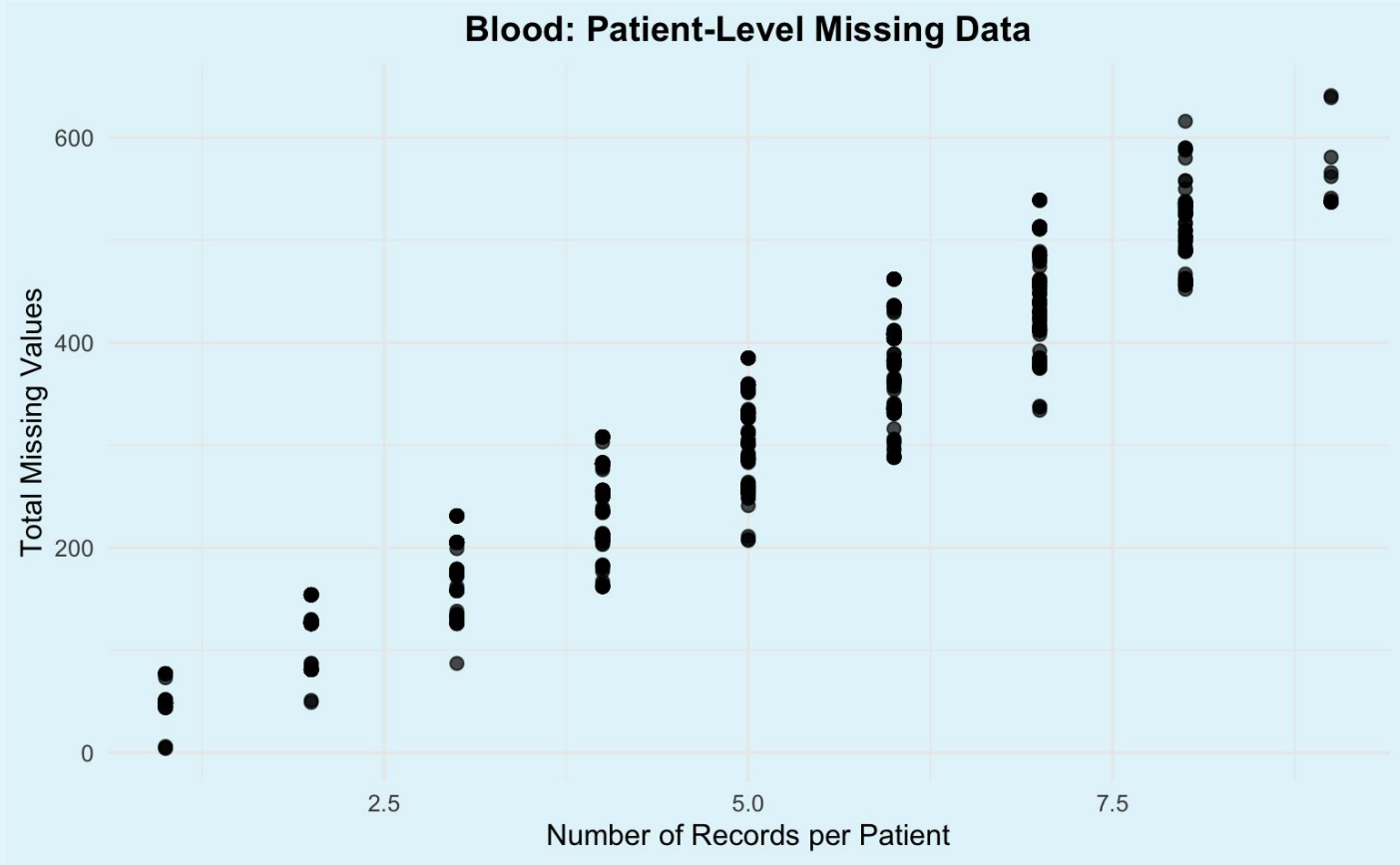
Blood Missing Data

Over 50 features missing a high percentage of data and many repeated values.

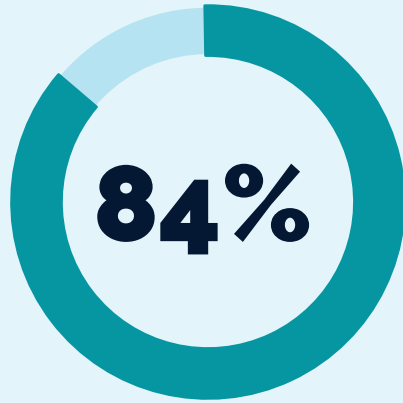


Blood Missing Data

As expected
having more
visits is
correlated with
an increase in
missing values.



Diet Missing Data



Every single diet feature is 84.6% missing skewing the data heavily

id_event

id

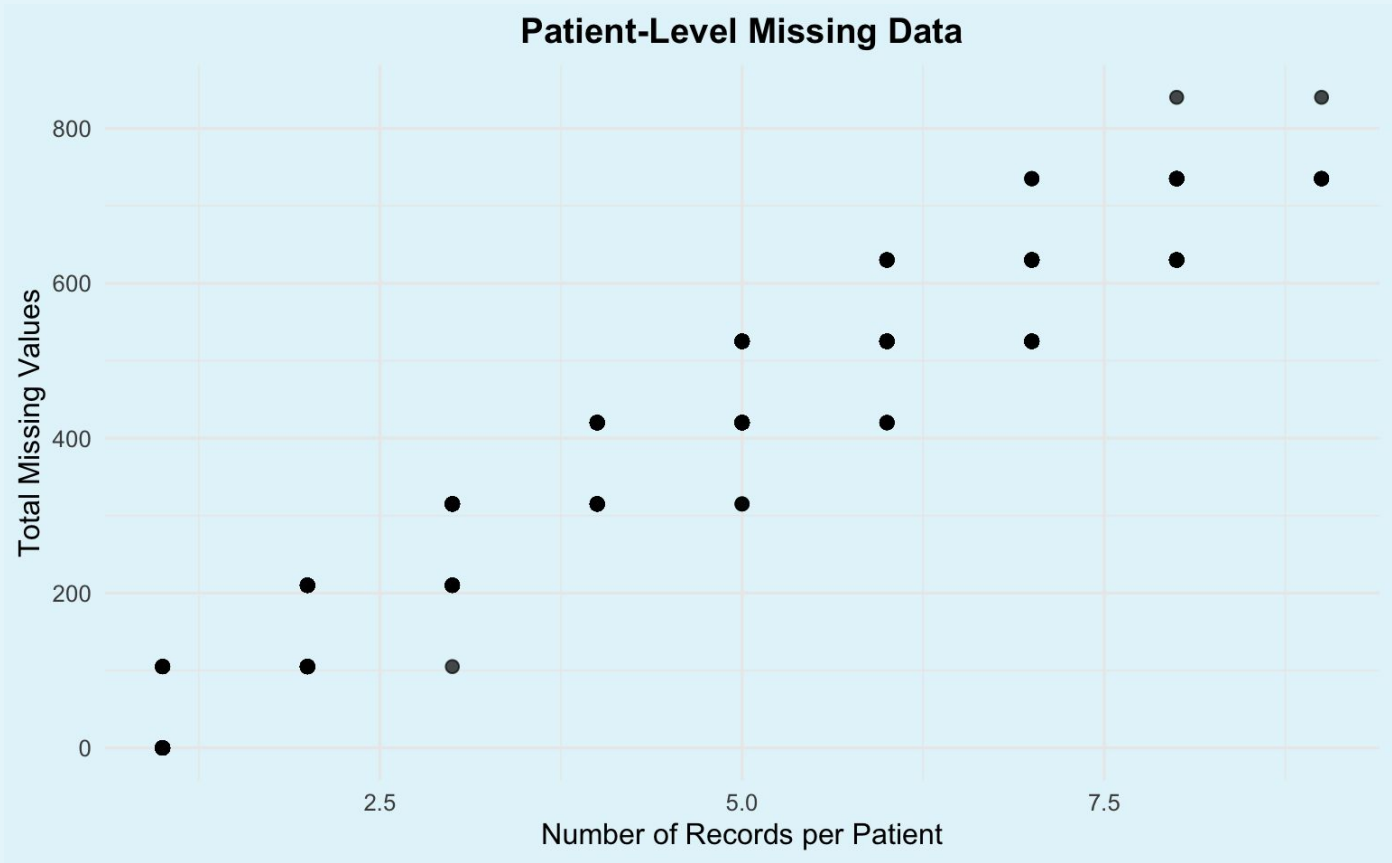
event

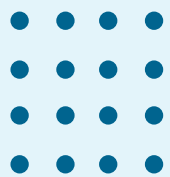
study

All of the classification features have a 0% missingness rate.

Diet Missing Data

Once again
having more
visits is
correlated with
an increase in
missing values.





02

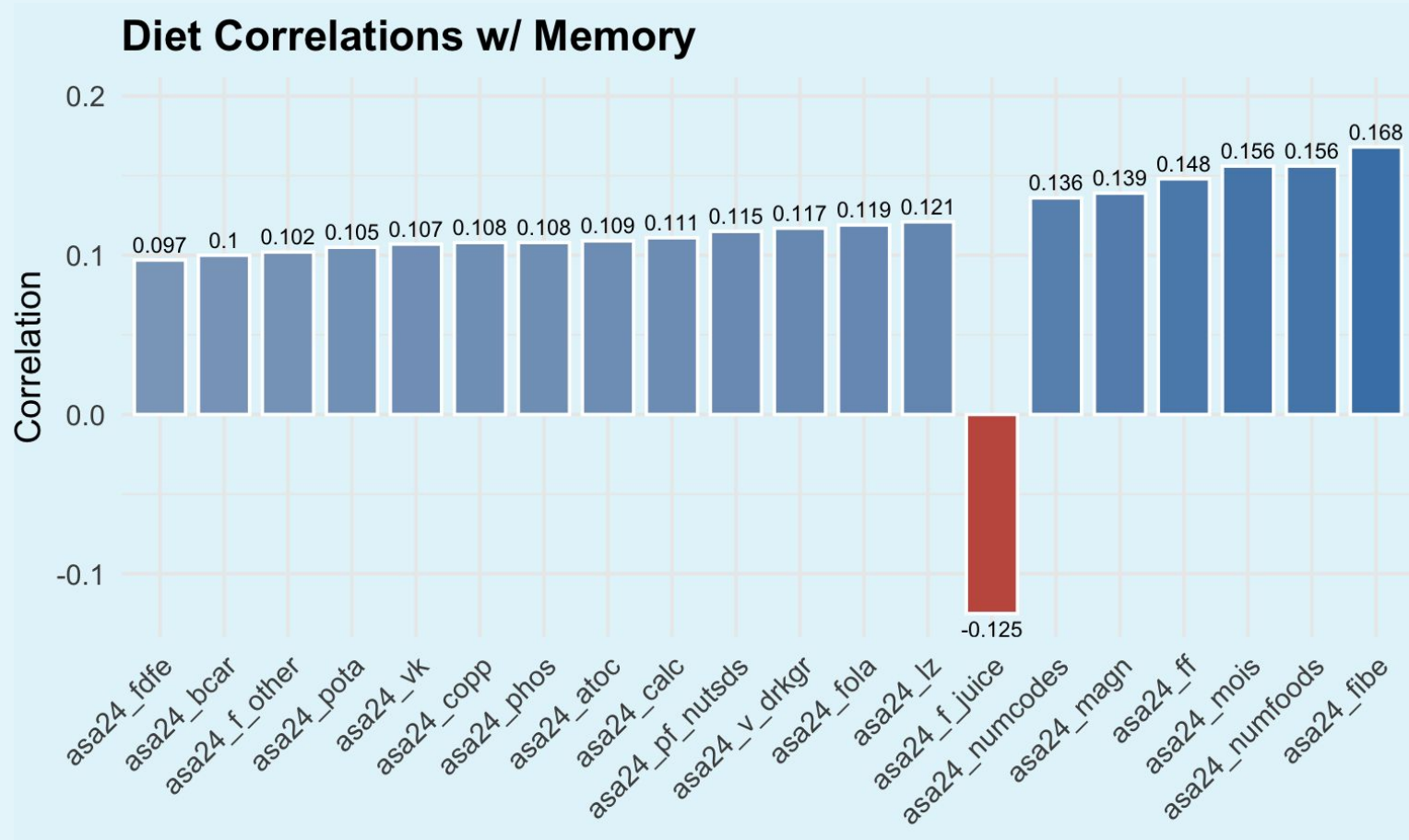
Cross- Sectional Model

With the information about the missing data we carefully select variables correlated with memory and put them through AIC to maximize accuracy while minimizing complexity.



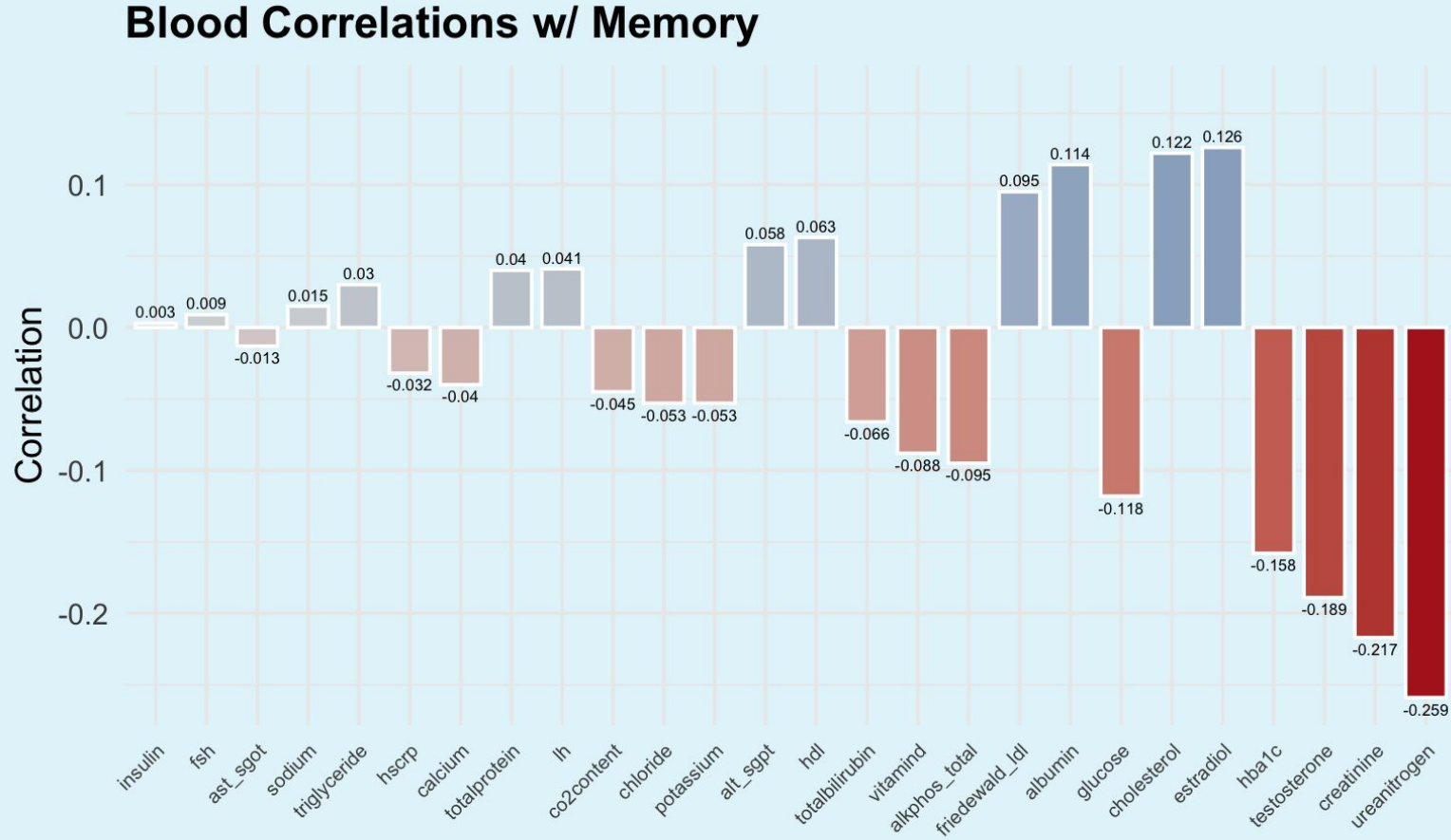
Diet Feature Selection

To reduce model complexity we only included features with the highest correlation to memory.



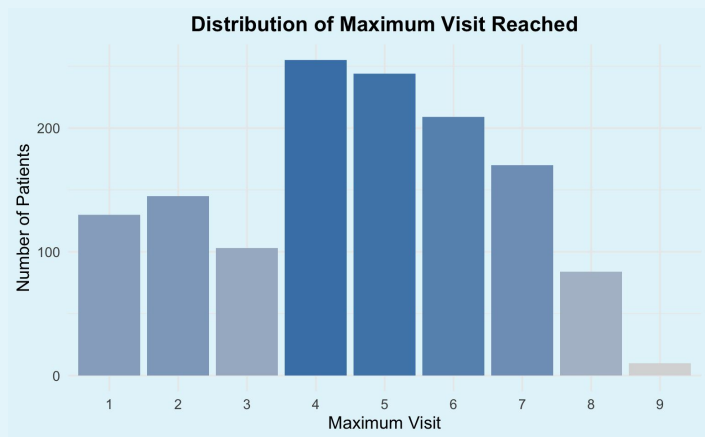
Blood Feature Selection

Features with less missingness were prioritized. To reduce model complexity we only included features with the highest correlation to memory.



Additional Feature Selection

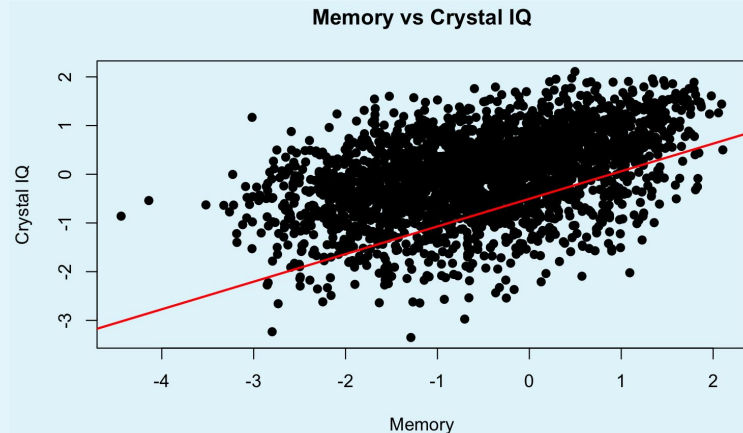
`n_visits`, `years_followed`,
`completed_v4`, `completed_f3`,



Features to capture confounding effects of many visits.

Age and Gender
Features directly requested by Client.

Crystal IQ



Highly correlated predictor of Memory.

Linear Mix Model

Stepwise AIC is a process that balances the complexity and accuracy of a model. It goes through a model and removes features that provide minimal insight into the target factor. We input the highly correlated Blood and Diet features along with the Additional features and then further break reduce the number of predictors.

26  9

AIC Reduced the Number of Predictors from 26 to 9.

Remaining Factors

Factor	P-Value	Estimate
- age	2e-16	-0.026125
- sexM	2e-16	-0.477937
- CrystIQ	2e-16	0.519937
- hba1c	0.0156	-0.101033
- insulin	0.0093	0.007689
- asa24_fibe	7.93e-05	0.008220
- n_visits	0.0778	0.048320
- years_followed	0.0903	-0.039671
- completed_f3TRUE	0.1221	0.095800

Adj. R²
0.4856

48.5% of the variance in memory can be explained by these 9 features.

GAM Model

Through Stepwise AIC we have now obtained the factors that give us the biggest bang for our buck in terms of complexity. We can now further drill down and make a more accurate model. Unlike LMM, GAM does not assume linearity meaning it can better capture non-linear relationships.

Linear Factors

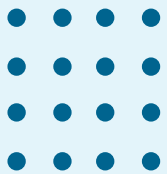
	Factor	P-Value	Estimate
-	sexM	2e-16	-0.49300
-	years_followed	0.1475	0.1475
-	n_visits	.0648	0.05030

Non-Linear Factors

	Factor	P-Value	EDF
-	age	2e-16	1.000
-	CrystlQ	2e-16	2.626
-	hba1c	0.0209	1.000
-	insulin	0.0133	1.777
-	asa24_fib	9.27e-05	1.000

Adj. R²
.498

49.8% of the variance in memory can be explained by these 8 features.

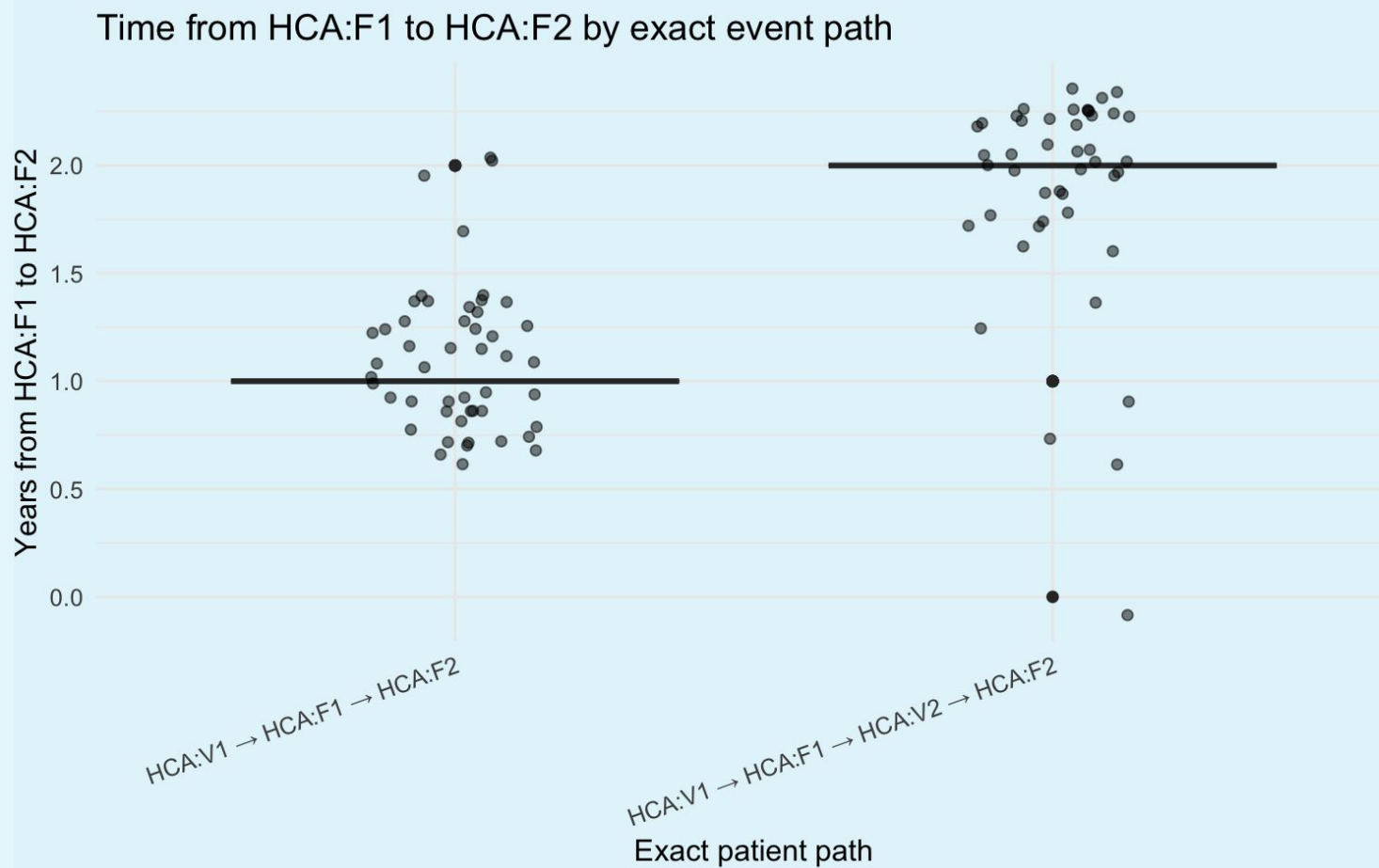


03

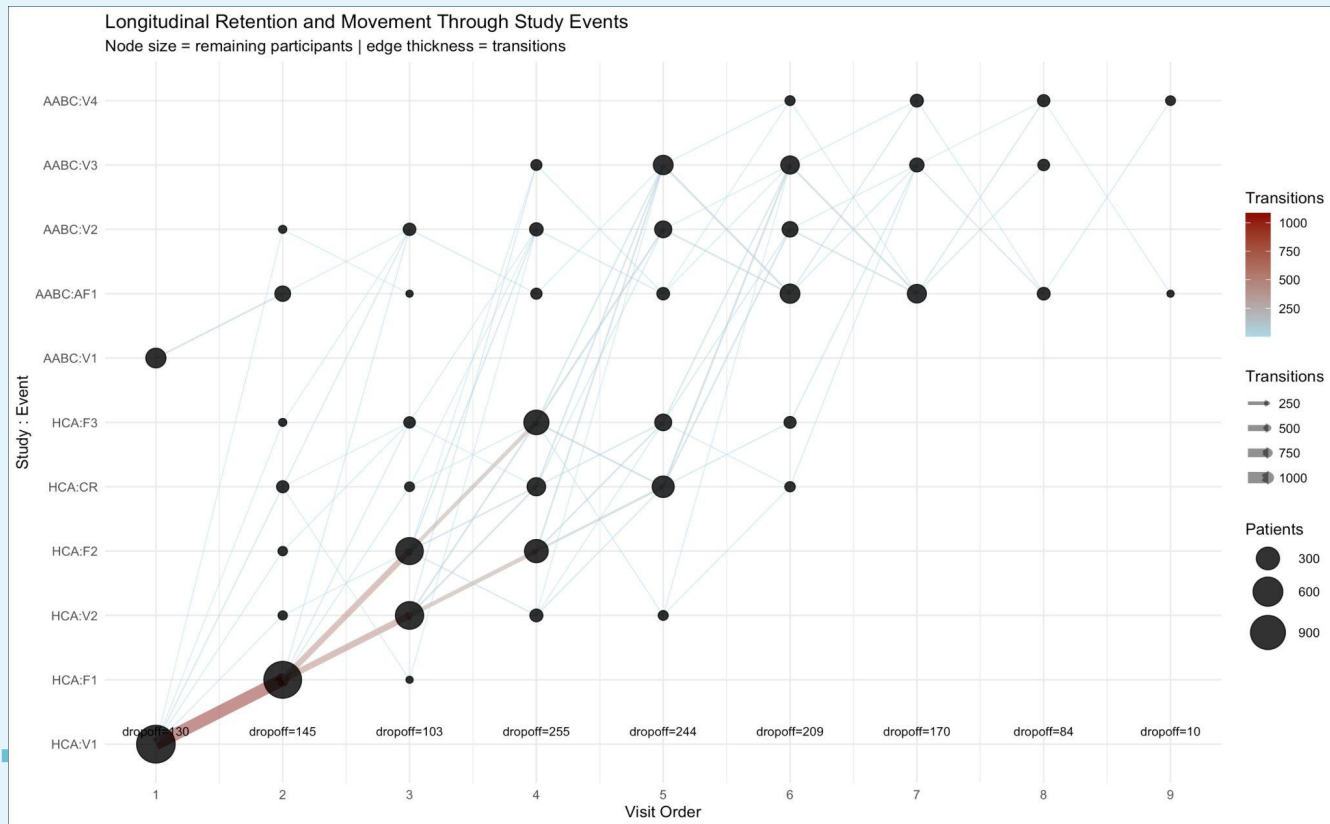
Patient Path



Patient Path

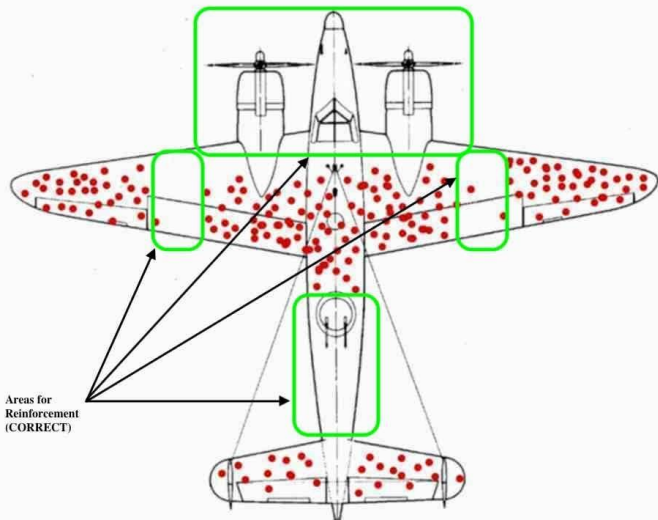


Retention Plot All Data



Survivor Bias

- Participants who remain in the study are likely healthier and more cognitively resilient than those who drop out
- Diet and biomarker effects may be biased if participants completing surveys/blood draws differ systematically from missing participants
- Missingness is likely informative (not random), meaning dropout itself may be related to health and cognition
- Model results may reflect predictors of continued participation rather than true population-level aging effects

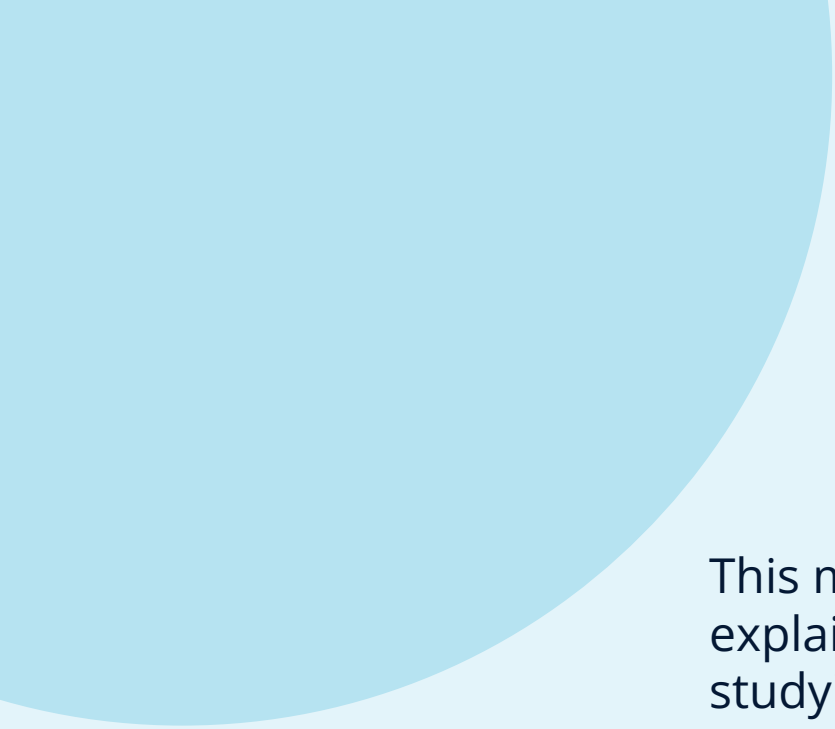


sex	Percent
F	56.9
M	43.1

MDL Sex Distribution	
Sex	Percent
F	56.1
M	43.9

Race	Percent
American Indian/Alaska Native	0.3
Asian	6.4
Black or African American	15.1
Hawaiian or Pacific Islander	0.2
More than one race	3.4
Unknown or not reported	1.9
White	72.7

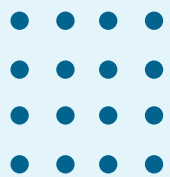
MDL Race Distribution	
Race	Percent
American Indian/Alaska Native	0.3
Asian	6.6
Black or African American	19.1
Hawaiian or Pacific Islander	0.1
More than one race	2.9
Unknown or not reported	1.9
White	69.1



ICC: 0.812

This means that 81.2% of Variation in memory is explained by who the patient is meaning that studying variation across a patient's multiple visits is the best way to determine the underlying effects of the factors.





04

Longitudinal Model

We have seen that a large portion of variance lies within a patients journey so to cut out the noise of cross patient variance we look at models the predict memory per patient journey.



Longitudinal LMM

Now that the model is looking at patients throughout their journey the additional factors are no longer necessary. Instead we can focus on the Blood, Dietary, and Classification factors see how they interact with memory.

Factors

- age 2e-16
- sexM 1.36e-09
- hba1c 0.033
- **insulin** 0.936
- asa24_fibe 1.13e-05

Insulin is revealed to be essentially completely uncorrelated with memory in the longitudinal model.

Likelihood Test

Mixed-Effects Model Comparison

Model	Fixed_Effects	Parameters	AIC	BIC	LogLik	Chisq	Df	p_value
Reduced Model	Age + Sex + HbA1c + Fiber Intake	7	2,028.3000	2,061.1000	-1,007.2000	NA	NA	NA
Full Model	Age + Sex + HbA1c + Insulin + Fiber Intake	8	2,030.3000	2,067.7000	-1,007.2000	0.0066	1	0.9353

No Evidence that insulin helps predict memory at all.

Longitudinal GAM

We have seen before that some features perform better when modeled non-linearly so we use GAM again in order to capture the patterns that LMM cannot.

Linear Factors

- sexM 4.08e-13

Non-Linear Factors

- age 2e-16
- CrystIQ 2e-16
- hba1c 0.0755
- asa24_fib 0.0027

Rand. Eff

- patient

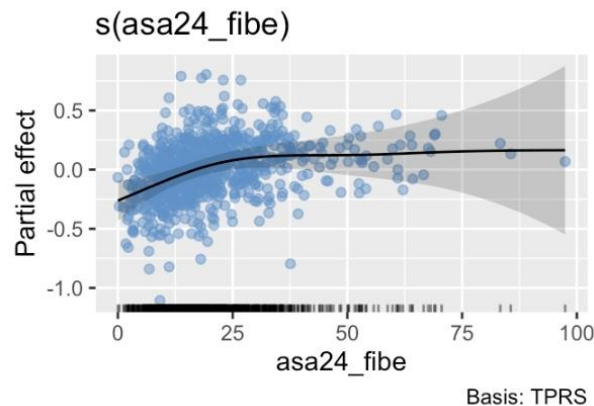
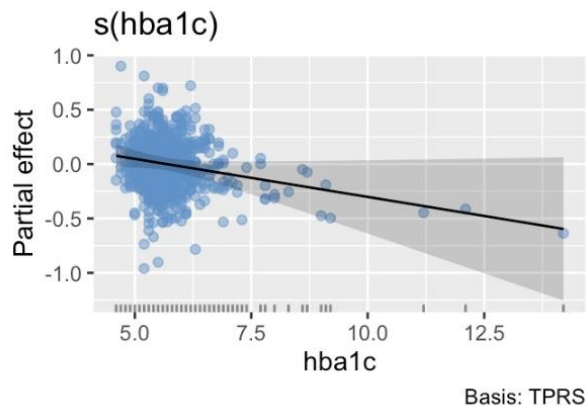
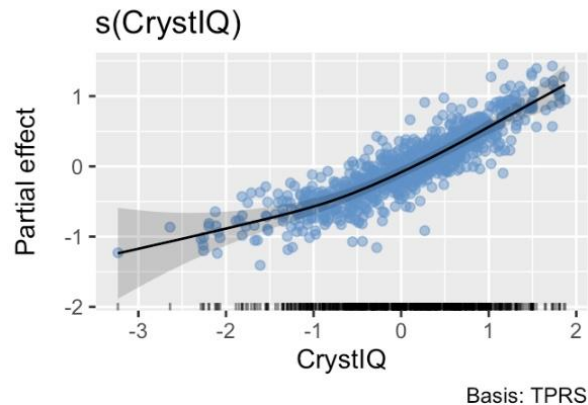
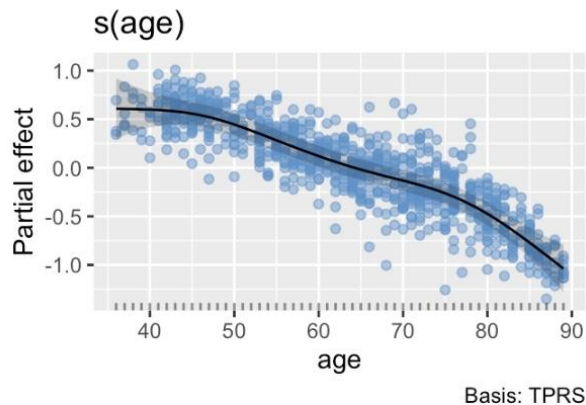
Adj. R^2
.442

44.2% of the variance in memory within patients can be explained by these 5 features.

Non-Linear Smoothing

```
memory_gamm <-
```

```
gamm(  
  Memory ~  
    sex +  
    s(age) +  
    s(CrystIQ) +  
    s(hba1c) +  
    s(asa24_fibe),  
  random =  
    list(  
      patient = ~1  
    ),  
  data = model_df,  
  method = "REML"  
)
```



Longitudinal GAM and Dropoff

We added back in the variables describing the patients so that the model may more easily capture the variance of patients journeys of various lengths and endpoints.

Linear Factors

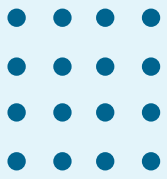
- sexM	4.08e-13
- n_visits	8.14e-05
- years_followed	0.0040
- completed_f3TRUE	0.2218
- completed_v4TRUE	0.0663

Non-Linear Factors

- age	2e-16
- CrystIQ	2e-16
- hba1c	0.10652
- asa24_fib	0.00405

Adj. R²
0.453

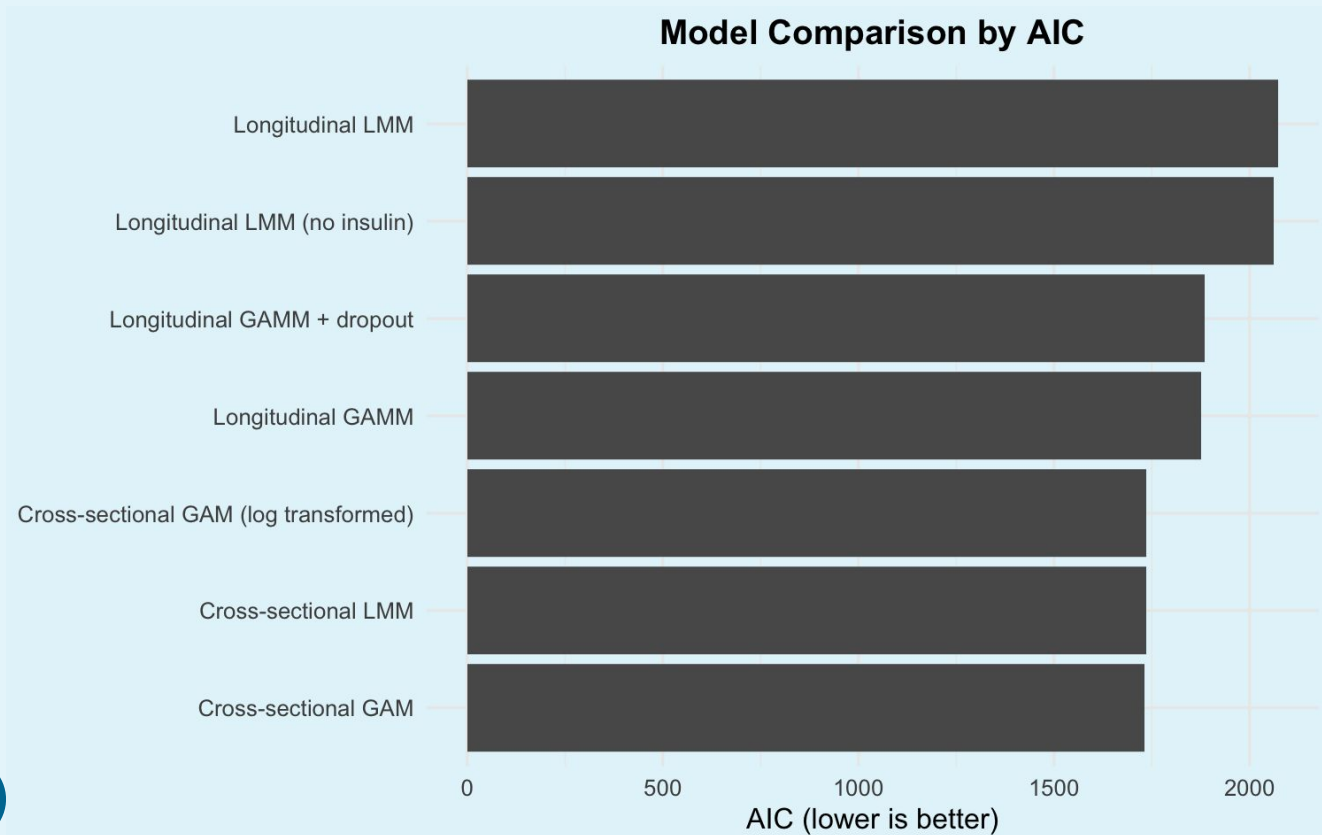
45.3% of the variance in memory within patients can be explained by these 9 features.



05

Conclusion

AIC Model Comparison



Results



The AABC/HCA analyses focused on whether modifiable lifestyle and metabolic factors are associated with cognitive aging, particularly Memory performance. The strongest and most consistent findings across models were:

- **Older age** was associated with lower Memory scores.
- **Higher HbA1c** (poorer glucose regulation) was associated with worse cognition.
- **Higher dietary fiber intake** was associated with better Memory performance.
- **Crystallized IQ** showed a strong protective relationship with cognition.
- Several relationships were **nonlinear**, meaning GAM/GAMM models captured the data substantially better than standard linear models.



Cross-sectional GAM models produced the best overall fit (AIC $\approx$ 1731), explaining nearly **49% of the variance** in Memory scores. Longitudinal mixed-effects models confirmed that repeated measurements within individuals were highly correlated (ICC $\approx$ 0.81), supporting the importance of participant-specific longitudinal modeling.

Challenges/Limitations



- HCA/AABC participants are relatively healthy, educated, and motivated, results may not generalize to the broader population at dementia risk.
- ASA24 captures one day of eating , a poor proxy for long-term dietary patterns
- Patients with more visits were initially counted more times, inflating correlations; we corrected this by averaging per patient, but a proper mixed-effects model is the right long-term solution



**Thank
You!**